Band-1 Computational Experiment: Map-Folding — Finite Exact Counts for Small Nets

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Band-1: Map-Folding / Stamp-Folding — Finite Exact Ledger + Box Certificates

Honesty stamp. This note does **not** claim that map-folding asymptotics are solved, that a closed-form formula exists, or that a polynomial-time counting algorithm is known. It does **not** claim any Clay / Millennium stamp or a finished Theory of Everything. **GATE-04 collection remains LOCKED** (out of scope). Band-1 here is a **literature ledger** of known exact map / stamp / meander / net counts for small sizes, plus **independent box certificates for small labeled $1 \times n$ stamp-folding counts only**. The next open rectangular net exact (notably 8×8) remains **OPEN**.

Collapse-list target: **1.09 — Map-folding exact counts for the next small nets**.

Abstract

Map folding and stamp folding ask how many distinct flat-folded stacks arise when a rectangular sheet (or a $1 \times n$ strip of stamps) is folded only along a fixed crease grid. Exact counts are classical hard enumeration problems (Koehler 1968; Lunnon 1968, 1971). Tonight's Band-1 probe:

1. Compiles a careful literature ledger (OEIS A000136 / A000682 / A001415 / A001418; Wikipedia *Map folding*; Sawada–Li 2012; Legendre 2014) with survey cutoff **2026-09-11**.
2. Independently box-certifies **OEIS A000136** for $n = 1, \ldots, 9$ by exhaustive permutation search with the same-parity crease non-crossing filter:

$$1,\ 2,\ 6,\ 16,\ 50,\ 144,\ 462,\ 1392,\ 4536.$$

3. Verifies on-box the relation $A000136(n) = n \cdot A000682(n)$ (semi-meanders) for the same range.
4. Names the **next open exact rectangular net**: $A001418(8)$ (8×8 map; literature known only for $n \leq 7$).

No new records beyond the classical small table are claimed.

Problem statement

Stamp folding ($1 \times n$). Labeled stamps $1, \ldots, n$ in a strip; folds only on creases between consecutive stamps; two foldings are distinct when their top-to-bottom stack permutations differ. Count = $A000136(n)$.

Map folding ($m \times n$). Rectangular crease grid; mountain/valley stacks of the unit cells. Square case $n \times n$ is OEIS A001418; known exacts only for $n \leq 7$.

Related. Semi-meanders $A000682$; blank (unlabeled) stamp shapes $A001011$; $2 \times n$ strips $A001415$.

Unsolved (explicitly not claimed here): closed form; poly-time algorithm; asymptotic connective constant to rigorous exactness; $A001418(8)$.

Literature ledger (survey 2026-09-11; large values cited, not re-proved)

Labeled $1 \times n$ stamp foldings — OEIS A000136

n	$A000136(n)$	Status
1	1	exact; **box-certified**
2	2	exact; **box-certified**
3	6	exact; **box-certified**
4	16	exact; **box-certified**
5	50	exact; **box-certified**
6	144	exact; **box-certified**
7	462	exact; **box-certified**
8	1392	exact; **box-certified**
9	4536	exact; **box-certified**
10	14060	exact (literature)
11	46310	exact (literature)
12	146376	exact (literature)

Sources: OEIS A000136; Koehler (1968); Lunnon (1968); Wikipedia *Map folding*.

Semi-meanders — OEIS A000682

Quotients $A000136(n)/n$: $1, 1, 2, 4, 10, 24, 66, 174, 504, ...$ (box-checked via n-divisibility for $n \leq 9$).

$2 \times n$ strips — OEIS A001415

$1, 1, 2, 8, 60, 320, 1980, 10512, 60788, ...$ (index $n = 0, 1, 2, ...$; Lunnon; extensions by Lunnon 2013 and later computational work). **Cited, not re-enumerated tonight.**

$n \times n$ maps — OEIS A001418

n	$A001418(n)$	Status
1	1	exact (literature)
2	8	exact (literature; classical)
3	1368	exact (literature)
4	300608	exact (literature)
5	186086600	exact (literature)
6	123912532224	exact (literature)
7	129950723279272	exact (literature)
8	—	**OPEN** (next small square net)

n	$A001418(n)$	Status

Source: Wikipedia *Map folding* / OEIS A001418 (survey 2026-09-11).

Method (box)

1. Represent a labeled $1 \times n$ folding as a permutation $p \in S_n$ (top to bottom).
2. Crease k joins stamps k and $k+1$. Two creases of the **same parity** cross if their four endpoints alternate in the linear order of p (patterns `ijij` / `jiji`).
3. Count permutations with **no** same-parity crossing (Koehler / Lunnon / Wikipedia characterization).
4. Fail-closed unit tests: forbid Wikipedia example 1324; accept all of S_3; accept identities.
5. Compare each $n \leq 9$ count to OEIS A000136; also check $A000136(n) = n \cdot A000682(n)$.

Code: `/workspace/mission-control/experiments/band1/map_folding_finite/verify_map_folding.py`.

Results (numbers!)

Unit tests

`PASS` (forbidden 1324; all S_3; identities $n = 1..6$).

Box-certified exacts (A000136)

n	Box count	Literature	Wall time (approx.)
1	**1**	1	$< 10^{-5}$ s
2	**2**	2	$< 10^{-5}$ s
3	**6**	6	$< 10^{-5}$ s
4	**16**	16	$< 10^{-4}$ s
5	**50**	50	$\sim 2 \times 10^{-4}$ s
6	**144**	144	~ 0.002 s
7	**462**	462	~ 0.014 s
8	**1392**	1392	~ 0.12 s
9	**4536**	4536	~ 1.2 s

All nine certificates completed on this box under the Band-1 time budget. Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`.

Literature next open (not solved here)

- $A001418(8)$ — exact number of foldings of an 8×8 **map** (first missing square-net exact; known table stops at $n = 7$).
- **Closed form / poly-time algorithm** for stamp or map folding — **OPEN**.
- Asymptotic growth rate estimates exist in the physics/combinatorics literature but are **not** claimed as theorems here.

Why Band-1 vs asymptotics

- **Band-1 (this note):** finite exact ledger + small exhaustive certificates; names the next open net without inventing values.
- **Not claimed:** connective-constant exactness; $A001418(8)$; formula for $A000136(n)$; complexity classification beyond citing the open status.

Non-claims (explicit)

- No Clay-closed stamp.
- No finished-ToE claim.
- **GATE-04 LOCKED** — no invented J; collection out of scope.
- Map-folding / meander **asymptotics not solved**.
- No claim that $n \times n$ table beyond $n = 7$ was computed tonight.
- No claim that $2 \times n$ (A001415) or blank-stamp (A001011) sequences were re-derived on box (literature only).

Reproducibility

```
python3 /workspace/mission-control/experiments/band1/map_folding_finite/verify_map_folding.py
```

Expect matching A000136 for $n = 1..9$ and JSON/SUMMARY write under the same directory.

References (selected)

1. Koehler, J. E. (1968), Folding a strip of stamps, *J. Combin. Theory* 5:135–152.
2. Lunnon, W. F. (1968), A map-folding problem, *Math. Comp.* 22:193–199.
3. Lunnon, W. F. (1971), Multi-dimensional map-folding, *Computer J.* 14:75–80.
4. Sawada, J. & Li, R. (2012), Stamp foldings, semi-meanders, and open meanders, *Electron. J. Combin.* 19(2):#P43.
5. Legendre, S. (2014), Foldings and meanders, *Australas. J. Combin.* 58:275–291 (arXiv:1302.2025).
6. OEIS A000136, A000682, A001415, A001418.
7. Wikipedia, *Map folding* (accessed 2026-09-11).

Paths

- Lab: `/workspace/mission-control/experiments/band1/map_folding_finite/`
- Paper: `/workspace/mission-control/papers/2026-09-11_Band1_map_folding_finite_probe.md`